

Bring Them Home

Flight Director • Lunar Return Mission

15 SHIFTS • 60 STOPS • GRADE 12 • BRING_THEM_HOME

THE OPENING CARD

Three people are in a spacecraft on the far side of a quarter-million miles, and something aboard has just exploded. You are the flight director, and the calls are yours to make. Power is falling. The cabin is cooling. The trajectory is drifting off the one path that gets them back, and the air they are breathing is already on a countdown. The crew can hear every word said on the loop.

The Failure

BEFORE THE SHIFT — GREET

Four tiers of consoles and one crew you have not met

Mission Control is one room in four tiers stepping down to the plot boards, and the teams are rows rather than offices. Reyes, the guidance lead, wants you known to every row before the shift, because a call on the loop is answered by whoever recognises the voice, and the room runs on exactly that.

PLAN CARD 110W

The bang was four minutes ago. Commander Hale reported it before any instrument did, which is the first thing that should worry you. Now three consoles hold numbers that contradict each other: one bus reads zero, the guidance computer has restarted with nothing loaded, and several cabin-pressure channels dropped together inside the same second. The integration lead, Evelyn Carter, is already saying that channels dropping together are one fault and not three. This shift you decide what is happening to the vehicle, get one agreed picture of it, and set how this room will work for the next five days. Hale, Ruiz and Sato can hear every word of this.

BRIEFING 29W

A crewed lunar spacecraft reports an explosion, falling electrical power, and contradictory telemetry. The player takes the flight-director console and must first determine what can still be measured reliably.

WHAT YOU DECIDE 14W

Create a trusted state vector and a prioritized anomaly list before commanding the crew.

1.1 What failed: the spacecraft or the sensors?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · TRACE

SCENE 37W

Nine minutes after the bang, three digital cabin-pressure channels fall together. A mechanical gauge, an acoustic leak monitor, and the sensor-electronics page are also live. The flight director wants one diagnosis before the crew starts opening panels.

GUIDE 58W

Open each pressure channel to see what it depends on. Keep the ones that would still stand if the sensor electronics were at fault, and untick the rest. Then name the shared source that moved the others. Three channels falling together is not three measurements agreeing; it is one measurement repeated three times if they share a supply.

ASK 14W

Which pressure evidence survives the dependency check, and what shared source moved the rest?

BEHIND THE BUTTON 155W

Why three digital channels can be one measurement. Redundant transducers usually share a power supply, a reference voltage and a signal conditioning card. Any of those failing moves all three readings the same way at the same moment, which is exactly what a real depressurisation looks like on the display.

What the other two instruments are for. A mechanical gauge has no electronics between the cabin and the needle. An acoustic monitor listens for escaping gas rather than measuring pressure at all. Neither depends on the suspect chain, which is why they carry more weight here than any number of digital channels.

Why the diagnosis has to come before the panels. Opening panels in a cabin that is genuinely losing air spends time the crew may not have. Sending them to chase an instrumentation fault costs the same hour with no leak to find. The flight director needs one answer, not a list of possibilities.

TAKES AS READ

several instruments can share one reference circuit · an independent measurement does not inherit that shared failure

VERDICT 74W

Agreement is strong evidence only when the measurements can fail independently. The three digital pressure channels all inherit the same reference circuit. When that reference shifts, all three move together even if cabin pressure does not. The mechanical gauge has a different physical path, so its steady reading still counts. The acoustic monitor also finds no leak. Tracing dependencies turns three alarming numbers into one electrical fault and preserves the evidence that still stands.

TAKEAWAY 16W

Several agreeing readouts count as one piece of evidence when they inherit the same physical reference.

1.2 Build the first state estimate

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · SEQUENCE

SCENE 34W

Three consoles hold contradictory numbers. One bus reads zero. The guidance platform has restarted with no record of what it saw. Nothing has been written down in an order the next controller could reconstruct.

GUIDE 64W

All four of these will be done, so this is not a list of options. Ask of each what would be lost or wasted if it came later. Telemetry across the failure is overwritten within minutes. A fit across mixed clocks produces a confident answer that is wrong. And a command is only worth issuing once the estimate it rests on carries an uncertainty.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a number without its units and its time is not a measurement

VERDICT 83W

A state estimate is only worth having if it survives being wrong, and the order is what makes that possible. Telemetry across the failure is overwritten within minutes, so it is frozen before anything is computed from it. Units, timestamps and calibration come next, because a fit across mixed clocks produces a confident answer that is wrong. Estimates then carry uncertainties rather than single values. And the commands that follow are the ones whose benefit holds across every state the data still allows.

TAKEAWAY 12W

A good emergency command works across uncertainty rather than assuming it away.

1.3 Spend the first ten minutes

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 37W

Four controllers, a crew flying a vehicle that is venting something, and ten minutes to the next tracking pass. Every subsystem wants the loop, and each believes its own problem is the one that kills the crew.

GUIDE 59W

All four options are real work that has to happen in the next hour. Ask of each what it costs in the only resource that is short, which is attention. Four controllers are already working one failure from four sets of numbers. Two versions of the same analysis can reach the crew, and they will fly whichever arrives first.

ASK 14W

Ten minutes to the next tracking pass. What does the flight director do first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

two rooms acting on different versions of one number is worse than either acting alone

VERDICT 78W

Attention is a physical resource in an emergency, just like propellant. The first decision determines where that resource goes. Four controllers are already working one failure from four sets of numbers. Without one log and one command authority, the same analysis can be repeated or contradicted. The crew may then execute whichever version reaches them first. Organising the room is not a delay before the technical work. It is what lets every later calculation and command count once.

TAKEAWAY 11W

Attention is a scarce physical resource during a fast-moving systems failure.

END OF THE SHIFT 17W

Use units, vectors, graphs, measurement uncertainty, common-mode failure as an evidence chain rather than as isolated facts.

Find the Spacecraft

BEFORE THE SHIFT — TRIAL

Walk the room, back tier to the boards

Everything in this room is arranged by how directly it touches the spacecraft, and the geography is the org chart. Walk it once before the shift. A question asked to the wrong row is a question that gets answered by somebody guessing rather than by the person holding the number.

PLAN CARD 102W

9 hours after the bang, and the station carrying the whole solution sets in 40 minutes. The guidance lead, Camila Reyes, will not sign the trajectory until she has one measurement that did not come through the same computer. Today you rebuild where the spacecraft is and how fast it is going, and buy the next observation. Fitting the measurements together is the job, and only one station is still up. Her trajectory officer, Dale Whitaker, wants a number now so the room can start planning the correction. Reyes wants the right number. They are both right, and there are 40 minutes.

BRIEFING 23W

Tracking data are intermittent, and the onboard computer has restarted. The team must reconstruct position and velocity from range, bearing, and timed observations.

WHAT YOU DECIDE 13W

Produce a consistent trajectory estimate and identify which new observation most reduces uncertainty.

2.1 Which measurement constrains what?

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · PROTOCOL

SCENE 39W

The onboard computer has restarted and the tracking is intermittent, so the state has to be rebuilt from whatever measurements can still be taken. The room has to say what each type buys before it schedules any of them.

GUIDE 67W

Four things the state estimate needs, and four ways of measuring. Pair them by asking what each measurement is physically sensitive to. An angle? A travel time? A frequency shift? Or a difference between two earlier answers? One of the four is built out of the others, which is why it is the noisiest. The next pass is intermittent, so what gets scheduled has to be chosen.

ASK 13W

Match each quantity the state estimate needs to the measurement that provides it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

describing where something is and how it is moving takes six numbers · waves: frequency, wavelength, speed — taken as read

VERDICT 70W

Each measurement constrains a different part of the spacecraft state. Direction comes from angles against known stars. Range comes from radio travel time. Doppler carries line-of-sight speed in the frequency shift. At 2 gigahertz, a closing speed of 1 kilometre per second shifts the carrier by about 7 kilohertz. Repeated positions give velocity, and repeated velocities give acceleration. That last step is noisy because each subtraction carries measurement error forward.

TAKEAWAY 7W

Navigation is a reconstruction from partial projections.

2.2 Reconstruct the state vector

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · SEQUENCE

SCENE 35W

The computer restarted and lost its alignment. A range, two bearing angles, and a Doppler trace remain. They were recorded on different clocks, and the next tracking pass will test whatever state the room reconstructs.

GUIDE 63W

All four steps will happen, so ask what each one needs to be true already. A fit across two clocks can look clean and still be wrong, so the frames come first. Range, angle and Doppler constrain different parts of the motion and are fitted together. Residuals separate a missing force from noise. And the burn is committed at an hour, not now.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

measurements only combine when they share a frame and a clock

VERDICT 78W

A useful state estimate has to predict the next observation. First, every measurement is moved onto one coordinate system and one clock. Otherwise a clean fit can still be wrong. Range, angle, and Doppler are then fitted together because they constrain different parts of the motion. The residuals show what the model missed. A pattern suggests an unmodeled force; random scatter suggests noise. Finally, both the state and its uncertainty are propagated to the time of the burn.

TAKEAWAY 12W

A useful state estimate predicts future observations and exposes its own failures.

2.3 What holds it in the curve

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · CHOICE

SCENE 35W

The free-return path swings the spacecraft round the Moon at 2,500 kilometres altitude. Somebody on the loop has asked why it does not simply fall onto the surface, and the room owes them an answer.

GUIDE 61W

All four options offer something that holds the spacecraft up. Ask of each whether it names a real force. Two of them invent one: an outward push, and lift with nothing to push against. Gravity is the only force acting here, and it points inward. So the question is what the sideways speed does to a path that is already falling.

ASK 14W

Why does the spacecraft need any speed at all to stay in that orbit?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an object moving in a circle is accelerating even at constant speed · kinematics — taken as read · newton's laws and free-body reasoning — taken as read

VERDICT 76W

Circular motion requires acceleration toward the centre. Around the Moon, gravity supplies that inward acceleration. The spacecraft is therefore falling during the entire orbit. Its sideways speed makes the falling path curve around the Moon instead of into the surface. Too little speed makes the path intersect the Moon. Too much opens the path outward. No engine must hold the spacecraft up. The crew feel weightless because there is no support force opposing the free fall.

TAKEAWAY 18W

A curved path needs a force toward the centre, and orbital motion is that force being exactly enough.

END OF THE SHIFT 18W

Use position, velocity, acceleration, coordinate systems, graphs of motion as an evidence chain rather than as isolated facts.

The Wrong Trajectory

BEFORE THE SHIFT — FOLLOW

The handover walk, done once at the pace it happens

A shift handover moves along the tiers in a fixed order, and each console hands over a state rather than a summary. Stay with whoever is doing it. The stops are where the state actually changes hands, and skipping one is how a shift starts with an assumption.

PLAN CARD 100W

18 hours in, the drift is confirmed, and this is where Whitaker and Carter stop agreeing for the rest of the mission. The path misses the entry corridor by 400 kilometres. Whitaker wants a modest burn this shift, arguing that a correction gets cheaper the earlier it is made. Carter wants to wait for the next tracking pass, because correcting a number you have not measured properly is how a small error becomes a committed one. This shift you settle it. That is Whitaker's whole case, and this shift it is the better one. There is no more propellant coming.

BRIEFING 31W

The spacecraft is drifting away from its planned free-return path. The player must use Newtonian gravity and impulse to determine whether a small correction now can prevent a large correction later.

WHAT YOU DECIDE 14W

Choose a correction that restores a safe return while preserving fuel and engine options.

3.1 From force to trajectory change

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · SEQUENCE

SCENE 37W

The spacecraft is drifting off the free-return path by an amount that is small today and fatal at the atmosphere. The crew wants to know why a few seconds of thrust now beats minutes of thrust tomorrow.

GUIDE 59W

These four are one chain, so each card has something before it and something after. Ask what has to exist first. A force before an acceleration. An acceleration before a change in velocity. The last card is why any of it matters. The changed velocity is carried for hours before arrival, which is why seconds now beat minutes tomorrow.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

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TAKES AS READ

a force acting on a mass produces an acceleration · kinematics — taken as read

VERDICT 78W

The chain starts with force applied for a measured time. Force divided by spacecraft mass gives acceleration. That acceleration changes velocity during the burn. The impulse, force times time, equals the change in momentum. For a fixed spacecraft mass, that sets the change in speed. The new velocity then changes where the spacecraft will be hours later. This is why an early correction can be small: the vehicle carries the changed velocity for a long time before arrival.

TAKEAWAY 11W

Trajectory control acts on velocity now to change position much later.

3.2 Small burn, large consequence

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · BALLPARK

SCENE 29W

A 30,000 kg spacecraft takes 6,000 N of thrust for 20 seconds. The flight-dynamics officer needs the resulting acceleration and change in speed before the correction burn is approved.

GUIDE 61W

Five numbers, and two of them belong to other questions. Surface gravity and specific impulse are not in this chain. Ask of each whether it is a force, a mass, or a time. And note the scale of the answer. Four metres per second looks like nothing beside an orbital speed. Days out, it moves the arrival by hundreds of kilometres.

ASK 5W

Estimate the change in speed.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

force applied over a time is an impulse · newton's laws and free-body reasoning — taken as read

VERDICT 79W

Force divided by mass gives an acceleration of 0.20 metres per second squared. Over 20 seconds, that produces a 4 metre-per-second change in speed. The same result comes from impulse divided by mass. Four metres per second looks tiny beside an orbital speed measured in kilometres per second. Applied days before arrival, however, it shifts the encounter by hundreds of kilometres. A late correction must produce the same positional change in far less time, so it costs much more.

TAKEAWAY 10W

Orbital corrections often rely on small velocity changes applied early.

3.3 What the pressure difference is for

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · CHOICE

SCENE 34W

The oxygen tank reads 40 psi, the cabin regulator is set to 15, and the flow has fallen to half of what the plan assumed. The line is the same line it was yesterday.

GUIDE 62W

All four options explain a flow that has halved. They differ in what they hold responsible: the difference, the path, the sensor, or the temperature. Ask of each whether the line is the same line it was yesterday. A pressure difference drives the flow, and how restrictive the path is decides the rate. A high-pressure tank delivers nothing through a blocked line.

ASK 15W

The tank reads 40 psi and the regulator wants 15. What is the flow doing?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

gas flows from higher pressure to lower pressure · newton's laws and free-body reasoning — taken as read

VERDICT 82W

Gas moves when a pressure difference exists across a path. The flow rate also depends on how restrictive that path is. A 40 psi tank feeding a 15 psi regulator still has a strong driving difference. If the measured flow falls by half, either that difference changed or resistance in the path increased. A partly closed valve, ice, or a loaded filter can do that. Tank pressure alone never guarantees flow, because even a high-pressure tank delivers nothing through a blocked line.

TAKEAWAY 14W

Something moves through a system for one reason and is held back for another.

END OF THE SHIFT 17W

Use Newton's laws, gravity, orbits, impulse, trajectory sensitivity as an evidence chain rather than as isolated facts.

Turn Without Wasting Fuel

PLAN CARD 99W

26 hours in. The guidance platform is off to save power, 1 thruster quad is dead, and Hale has to turn the spacecraft 90 degrees by hand while Sato reads angles aloud.

Kwame Mensah, who owns the structural dynamics, has done the sums and does not like them. This shift you agree what the rotation numbers mean, write the procedure, and work out what a thruster actually does. Every degree of turn is paid for twice, once to start it and once to stop it. Let it tumble and the antenna, the array and the burn attitude go together.

BRIEFING 20W

The main guidance platform is unavailable, and the crew must reorient the spacecraft using limited thrusters and visible reference objects.

WHAT YOU DECIDE 13W

Develop a controlled reorientation procedure that stops the rotation at the required attitude.

4.1 Read rotational motion

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · PROTOCOL

SCENE 39W

The main guidance platform is unavailable and the crew is about to fly the vehicle by hand. Before anybody touches a thruster the room states the rotational mechanics plainly, because each one has a consequence the crew will feel.

GUIDE 67W

Four changes on the left, and four consequences on the right. Pair them by asking what each change does to the two terms that matter: the torque, and the resistance to turning. One of the four adds no external torque at all, which is what makes it different from the others. The crew is about to fly this by hand and will feel every one of them.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a torque is what makes something start or stop turning · newton's laws and free-body reasoning — taken as read

VERDICT 82W

Attitude control is Newtonian mechanics about a centre of mass. Torque grows with force and with the lever arm from that centre. The same torque produces less angular acceleration when the moment of inertia is larger. Equal and opposite torques applied for equal times cancel the net angular impulse. Internal mass motion can change how fast different parts rotate, but it cannot change the vehicle's total

angular momentum without an external torque. Those four rules set the logic of every manual turn.

TAKEAWAY 10W

Attitude control is Newtonian mechanics around a center of mass.

4.2 Execute a manual attitude maneuver

MISSION INTEGRATION • FLIGHT DIRECTOR • A A PLACE • FLY

SCENE 30W

The guidance platform is off. The crew must turn the spacecraft 90 degrees using 2 opposing thruster pairs and a window reticle. Every extra correction spends propellant reserved for entry.

GUIDE 55W

You set two things: how hard to start the turn and where to start braking. Then the plan runs and you watch it. Nothing damps this rotation, so whatever you start you must stop deliberately, and the braking has to lead the target rather than meet it. Every extra correction spends propellant reserved for entry.

ASK 19W

How hard do you start the turn, and where do you begin braking so the spacecraft arrives at rest?

BEHIND THE BUTTON 153W

Why the brake has to lead. With nothing to slow the spacecraft, the rotation continues at whatever rate you gave it until an equal impulse takes it away. Firing at the target means arriving at the target still turning, and then correcting past it — which is where the propellant goes.

What the reticle actually gives you. A mark on the window and a star or a horizon is enough to see the attitude, not the rate. Rate has to be inferred from how fast the view moves, which is why a small starting impulse is easier to fly than a large one even though it takes longer.

Why the propellant matters more than the minutes. What is being spent belongs to the entry budget. A turn flown in half the time with three corrections can cost the margin that the atmosphere entry needs, and there is no way to get it back.

TAKES AS READ

a torque changes angular speed • rotation continues after a thruster stops firing • newton's laws and free-body reasoning — taken as read • rotational kinematics — taken as read

VERDICT 81W

A thruster pulse changes angular speed; it does not create a turn that stops by itself. After the opening pulse ends, the spacecraft keeps rotating at the new rate. The opposite thruster needs time and angle to remove that rate. For equal start and stop pulses, the braking distance is the same kind of problem as stopping a moving cart. A late brake therefore overshoots. The maneuver is successful only when both angle and angular rate are small at the target.

TAKEAWAY 17W

In undamped rotation, stopping the torque does not stop the motion; braking must begin before the target.

4.3 Torque from a thruster

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · BALLPARK

SCENE 31W

A 200 N thruster fires at right angles, 3 metres from the centre of mass. The crew needs the torque before it can predict how fast the vehicle starts to turn.

GUIDE 61W

Five numbers, and three of them belong elsewhere: the vehicle mass, the docking ring, and a cosine. Ask of each whether the turning effect depends on it. The force is at right angles here, so no angle factor is needed. And note what the crew can actually change. Choosing a different thruster pair changes the lever arm without changing the thruster.

ASK 8W

Estimate the torque about the centre of mass.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a longer lever arm gives the same force more turning effect · newton's laws and free-body reasoning — taken as read · rotational kinematics — taken as read

VERDICT 78W

Torque is force times lever arm times the sine of the angle between them. Here the force is perpendicular, so that sine is one. The calculation is therefore just 200 newtons times 3 metres. The result is 600 newton metres. Move the same thruster closer to the centre of mass and the torque falls in direct proportion. On this vehicle, the crew can choose which thruster pair fires. That choice changes the lever arm without changing the thruster.

TAKEAWAY 11W

Force location matters as much as force magnitude in rotational control.

4.4 Off-nominal, or off-scale

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · BELT

SCENE 36W

The console has flagged more parameters than the team can look at before handover. Some of them are the spacecraft behaving unexpectedly. The rest are the measurement failing, and the two look identical in a list.

GUIDE 50W

Two bins. A spacecraft doing something unexpected moves in ways that are physically possible and that other parameters agree with. A measurement failing moves in ways nothing else corroborates — instantly, off the end of the scale, or to exactly zero. Sort on whether the rest of the vehicle agrees.

ASK 17W

Send each flag to the bin that says whether the vehicle or the sensor is the problem.

BEHIND THE BUTTON 93W

Why this is the first question, always. The response to the spacecraft misbehaving and the response to an instrument misbehaving are opposite: one is an intervention, the other is ignoring a reading and fixing a sensor. Getting it backwards spends a consumable answering something that never happened.

What a failed sensor looks like. A step no physical process could produce, a value past the end of the range, a reading that has not changed in an hour when everything around it has, or a number that agrees with nothing else on the vehicle.

TAKES AS READ

a reading comes from a sensor, and a sensor can fail

VERDICT 165W

Every flag on this board is a number that moved, and the number is not the thing. A spacecraft that is genuinely warming shows it in more than one place: the structure temperature moves with the radiator outlet, the power draw agrees, and the rate is one a physical process could produce. A sensor that has failed shows none of that. It steps instantly, or parks past the end of its range, or freezes at exactly the same value while everything around it changes, or contradicts two other measurements of the same thing. That is why corroboration is the test rather than plausibility — a failed sensor can produce an entirely plausible number. And the cost of confusing them is asymmetric. Answering a sensor failure with an intervention spends propellant or power on a problem that does not exist, and treating a real excursion as a bad reading is how a crew arrives at the point where there is nothing left to do about it.

TAKEAWAY 15W

The vehicle and the measurement fail differently, and only one of them needs an intervention.

4.5 What the pressure difference is for

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · CALLBACK · CHOICE

SCENE 34W

The oxygen tank reads 40 psi, the cabin regulator is set to 15, and the flow has fallen to half of what the plan assumed. The line is the same line it was yesterday.

GUIDE 62W

All four options explain a flow that has halved. They differ in what they hold responsible: the difference, the path, the sensor, or the temperature. Ask of each whether the line is the same line it was yesterday. A pressure difference drives the flow, and how restrictive the path is decides the rate. A high-pressure tank delivers nothing through a blocked line.

ASK 15W

The tank reads 40 psi and the regulator wants 15. What is the flow doing?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

gas flows from higher pressure to lower pressure · newton's laws and free-body reasoning — taken as read

VERDICT 82W

Gas moves when a pressure difference exists across a path. The flow rate also depends on how restrictive that path is. A 40 psi tank feeding a 15 psi regulator still has a strong driving difference. If the measured flow falls by half, either that difference changed or resistance in the path increased. A partly closed valve, ice, or a loaded filter can do that. Tank pressure alone never guarantees flow, because even a high-pressure tank delivers nothing through a blocked line.

TAKEAWAY 14W

Something moves through a system for one reason and is held back for another.

END OF THE SHIFT 21W

Use torque, angular acceleration, moment of inertia, angular momentum, center of mass as an evidence chain rather than as isolated facts.

The Power Budget

PLAN CARD 110W

34 hours in, and the argument changes shape. Priya Shah, who runs the power systems, has the reserve down to hours and wants everything non-essential off by the end of the shift. Aaron Brooks, on thermal and life support, says the two things on his list are the scrubber fan and the cabin heaters, and that his crew are already sleeping in everything they own. The bus is sagging at 24 volts instead of 28 while the current sits near where it should be, which is Shah's other problem. This shift you work out how long the batteries last, decide why the voltage is low, and choose what goes off.

BRIEFING 19W

Fuel cells and batteries are degraded. Every electrical load now competes with guidance, communications, thermal control, and life support.

WHAT YOU DECIDE 13W

Build a load-shedding plan that preserves essential functions through the next critical maneuver.

5.1 How long can the battery last?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · BALLPARK

SCENE 31W

The fuel cells are gone and the crew is on entry batteries days earlier than any plan allowed for. The flight surgeon wants an endurance number before the next powerdown decision.

GUIDE 68W

Five numbers, and three of them belong to other questions: the bus voltage, the brief peak, and the time to the next burn. Ask of each whether it is an amount of energy or a rate of use. Energy divided by a rate leaves a time. And note what the answer assumes: a flat load, and every last watt-hour available. A cold battery gives less than its rating.

ASK 4W

Estimate the ideal endurance.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

energy is a stock, and power is the rate it is drawn at · circuits: current, voltage, resistance, Ohm's law — taken as read

VERDICT 81W

Energy and power are different things, and the division between them is the whole calculation. An inventory in kilowatt-hours divided by a draw in kilowatts leaves hours. It is an ideal figure. It assumes the load stays flat and that every last watt-hour is available, and neither is true — a cold battery gives less than its rating. It is still the number every other decision this shift gets argued against, so it is worth having early and worth calling approximate.

TAKEAWAY 15W

Power is the rate of energy use; endurance depends on both energy inventory and load.

5.2 Why is bus voltage collapsing?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 30W

The power bus is sagging at 24 volts instead of 28 while the total load current sits near its planned value. Engineers want to start pulling equipment off the bus.

GUIDE 63W

Four options, and three readings have to be explained together. Generation is normal. Total load current is near plan. Something is hot. Ask of each option how many of those it covers. A failed array or a hidden load would move the current. A bad meter would not make anything warm. Equipment is about to be pulled off the bus on this answer.

ASK 10W

Which fault fits the voltage, current, generation, and thermal evidence?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

current through a resistance drops voltage and makes heat

VERDICT 82W

Voltage, current, and heat describe the same electrical fault from different angles. A resistive junction creates a voltage drop when current passes through it. The lost electrical power becomes heat at that junction. Here generation is normal and the total load current is near plan, so neither a failed array nor a hidden load fits. The hot connection and low downstream bus voltage point to resistance in the path. Measuring all three quantities localizes the fault instead of guessing from voltage alone.

TAKEAWAY 19W

Electrical faults are easier to localize when voltage, current, and heat are treated as parts of one energy picture.

5.3 Shed load without losing the mission

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · ALLOCATE

SCENE 32W

41 amp-hours remain for a 62-hour return. Current draw is 4.2 amps. Guidance, communications, thermal control, and life support all want power, and at least one load must be reduced within minutes.

GUIDE 62W

Forty-one amp-hours, sixty-two hours to fly, and a current draw of 4.2 amps that does not fit. Watch the questions list rather than the loads: it shows what the spacecraft can still do under the current plan. Every hour you buy for one load is an hour you have taken from another, and at least one has to come down within minutes.

ASK 10W

How many hours of each load can the spacecraft afford?

BEHIND THE BUTTON 135W

What an amp-hour buys. One amp for one hour. At 4.2 amps the whole reserve lasts under ten hours, against sixty-two to fly, so this is not a trimming exercise: the load has to fall by a factor of several and stay down.

Why the loads are not equal. Guidance and communications can be intermittent — used for a pass and switched off. Thermal control has a time constant of hours, so it can be cycled. Life support cannot be interrupted at all. That difference is what makes a plan possible.

Why minutes matter for the first cut. Every hour spent deciding is drawn at the current rate, so a slow decision has already spent part of what it was allocating. The first reduction buys the time to make the rest of the plan properly.

TAKES AS READ

amp-hours equal current multiplied by time · some loads can be duty-cycled while others must run continuously · newton's laws and free-body reasoning — taken as read

VERDICT 81W

Power shedding is not a ranking of important systems. It is an allocation of current over time. Life support must run through the return. Guidance needs power during specific navigation windows. Thermal control can be duty-cycled, but not removed, because cold batteries lose usable capacity. High-bandwidth communications is the flexible load. When runtime is converted into amp-hours, the required functions consume almost the entire inventory. The remaining energy buys only short communications windows, so continuous broadband has to be given up.

TAKEAWAY 20W

A finite energy budget is a set of tradeoffs between load rate, runtime, and the mission conditions each load protects.

5.4 Which measurement constrains what?

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · PROTOCOL

SCENE 39W

The onboard computer has restarted and the tracking is intermittent, so the state has to be rebuilt from whatever measurements can still be taken. The room has to say what each type buys before it schedules any of them.

GUIDE 67W

Four things the state estimate needs, and four ways of measuring. Pair them by asking what each measurement is physically sensitive to. An angle? A travel time? A frequency shift? Or a difference between two earlier answers? One of the four is built out of the others, which is why it is the noisiest. The next pass is intermittent, so what gets scheduled has to be chosen.

ASK 13W

Match each quantity the state estimate needs to the measurement that provides it.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

describing where something is and how it is moving takes six numbers · waves: frequency, wavelength, speed — taken as read

VERDICT 70W

Each measurement constrains a different part of the spacecraft state. Direction comes from angles against known stars. Range comes from radio travel time. Doppler carries line-of-sight speed in the frequency shift. At 2 gigahertz, a closing speed of 1 kilometre per second shifts the carrier by about 7 kilohertz. Repeated positions give velocity, and repeated velocities give acceleration. That last step is noisy because each subtraction carries measurement error forward.

TAKEAWAY 7W

Navigation is a reconstruction from partial projections.

END OF THE SHIFT 17W

Use current, voltage, resistance, electrical power, energy budget as an evidence chain rather than as isolated facts.

A Dangerous Battery Configuration

BEFORE THE SHIFT — HUNT

Six consumable logs, and the board reconciles four

Everything the crew has left is tracked in logs that update at different times, and two of them do not match the board. Until they are found and reconciled the room does not know how many days it has, and every plan on the wall is built on that number.

PLAN CARD 100W

42 hours in. The electrical load officer, Ana Ferreira, has found a battery module on the damaged bus that would buy 11 hours, and Shah wants it connected before the next pass. Nobody knows its state of charge, and the connectors were never meant for this. Ferreira has already reported one connector on that bus running warm and cannot say why. This shift you decide whether that configuration is safe and what has to be measured first. Shah is counting the hours and Brooks is counting the people. Sealed cabin. Three of them. 11 hours is not worth a fire.

BRIEFING 25W

Engineers propose reconnecting battery modules, but a mismatched module could overheat or reverse-charge. The player must reason through series, parallel, internal resistance, and fault isolation.

WHAT YOU DECIDE 13W

Approve only a configuration whose voltage, current sharing, and failure behavior are understood.

6.1 Series or parallel?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · PROTOCOL

SCENE 33W

Engineers want to reconnect battery modules to buy hours of endurance. The room agrees what each wiring choice does before anything is bolted up, because adding a module can also take capability away.

GUIDE 61W

Four wiring changes, and four consequences. Pair them by asking what stays the same in that arrangement: the voltage across each part, or the current through it. Then ask what one fault does. Adding a module can take capability away, because a single open cell can break a whole series path. This is being bolted up to buy hours of endurance.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a battery has both a voltage and a current it can supply

VERDICT 82W

Circuit topology determines voltage, current sharing, and fault behavior. Equal cells in series add voltage while carrying the same current. One open cell can therefore break the whole series path. Equal cells in parallel keep about the same voltage and share the current. That arrangement can add useful capacity. A resistive parallel branch carries less than its share, forcing the other branches to carry more.

Redundancy only helps when the connection and protection rules prevent one fault from defeating the whole arrangement.

TAKEAWAY 11W

Redundancy is useful only when connection and protection behavior are explicit.

6.2 Heating at a bad connection

POWER & CIRCUITS · POWER & CIRCUITS BAY · A PERSON · BALLPARK

SCENE 33W

A connector in the improvised battery path develops 0.05 ohms while carrying 20 amps. The joint is inside a sealed cabin. The power team needs its heating rate before the hardware is approved.

GUIDE 71W

Five numbers, and three of them belong elsewhere: the bus voltage, the whole cable run, and half the current. Ask of each whether the heating at this joint depends on it. Note the square: at half the current the same joint gives off a quarter of the power. And note where it is. Twenty watts is nothing on a radiator and a great deal in one connector in a sealed cabin.

ASK 7W

Estimate the heating power at the connection.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a resistance carrying current turns electrical power into heat

VERDICT 79W

Resistive heating is current squared times resistance, so the current term dominates. The same connector at half the current would give off a quarter of the power. The resistance sounds like nothing written down. The wattage it makes at this current is what decides whether the junction runs warm or starts a fire. 20 watts is nothing spread over a radiator. It is a great deal in one connector, in a sealed cabin, with no air moving over it.

TAKEAWAY 13W

Small resistances matter when current is large because heating scales with current squared.

6.3 Qualify the emergency battery plan

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · VERIFY

SCENE 38W

The emergency module reads 31 volts against a nominal 37. Its state of charge is uncertain, and the available connector is rated for 8 amps. The crew needs more endurance, but the hardware is inside the sealed cabin.

GUIDE 58W

Lock what current you expect before anything is connected. Then make the connection and decide whether to spend what it costs to measure the electrical effect. The connector is rated for 8 amps and the module reads 31 volts against a nominal 37, so the prediction is about whether this is safe as well as whether it helps.

ASK 16W

What current do you expect, and will you verify the electrical effect before qualifying the module?

BEHIND THE BUTTON 141W

Why the voltage matters more than it looks. A module at 31 volts against 37 nominal is either partly discharged or damaged, and those imply very different currents when it is connected to a healthy bus. The current is set by the difference between the two, divided by the resistance of everything between them.

What the connector rating means. Eight amps is what the hardware can carry continuously. Exceeding it inside a sealed cabin is not a component failure — it is heat, and possibly smoke, in the volume the crew is breathing.

Why measure rather than assume. The prediction can be right and the hardware still behave differently: a damaged cell, a high-resistance joint, a connector not fully seated. Spending the measurement is what converts a plan into a qualification, and not spending it leaves the crew relying on arithmetic.

TAKES AS READ

a voltage difference between connected batteries drives current · a current rating is a physical limit, not a suggestion

VERDICT 82W

Two batteries at different voltages exchange current as soon as a conductive path joins them. The connection can look mechanically perfect while the current is already above a connector rating. Protection limits the consequence of being wrong, but it does not tell you what happened. The electrical result still has to be measured. Here the test lead survives, yet the measured current is too high for the intended connector. That makes the direct configuration unacceptable even though the test itself completed normally.

TAKEAWAY 18W

A successful electrical action is not a verified outcome until the current and heating it produces are measured.

6.4 Reconstruct the state vector

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · SEQUENCE

SCENE 35W

The computer restarted and lost its alignment. A range, two bearing angles, and a Doppler trace remain. They were recorded on different clocks, and the next tracking pass will test whatever state the room reconstructs.

GUIDE 63W

All four steps will happen, so ask what each one needs to be true already. A fit across two clocks can look clean and still be wrong, so the frames come first. Range, angle and Doppler constrain different parts of the motion and are fitted together. Residuals separate a missing force from noise. And the burn is committed at an hour, not now.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

measurements only combine when they share a frame and a clock

VERDICT 78W

A useful state estimate has to predict the next observation. First, every measurement is moved onto one coordinate system and one clock. Otherwise a clean fit can still be wrong. Range, angle, and Doppler are then fitted together because they constrain different parts of the motion. The residuals show what the model missed. A pattern suggests an unmodeled force; random scatter suggests noise. Finally, both the state and its uncertainty are propagated to the time of the burn.

TAKEAWAY 12W

A useful state estimate predicts future observations and exposes its own failures.

Use series circuits, parallel circuits, internal resistance, Joule heating, fault isolation as an evidence chain rather than as isolated facts.

The Cabin Is Cooling

PLAN CARD 118W

Two days after the bang, Brooks wins one, and it costs Shah her margin. The cabin is at four degrees and falling, condensation is running down the walls, and Ruiz has stopped being able to write legibly. Today you work out how fast the vehicle is losing heat, decide which way it is leaving, and choose what to insulate. There is power to fight it one way and not three. Average temperature is also the wrong number: batteries fail at their own local temperature, and the coldest spot in that cabin is the wall behind the pack the entry sequence depends on. Brooks is arguing for the crew, Shah for the batteries, and they want the same watts.

BRIEFING 23W

With most systems powered down, cabin temperature falls and condensation forms. The player must balance conduction, convection, radiation, metabolic heat, and battery performance.

WHAT YOU DECIDE 14W

Create a thermal survival plan that protects crew, electronics, and batteries with minimal power.

7.1 How much does the cabin cool?

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · BALLPARK

SCENE 34W

With the systems powered down the cabin is losing about a kilowatt more than it generates. The flight surgeon wants to know how long the crew has before the temperature becomes a medical problem.

GUIDE 64W

Five numbers, and two of them are the same three hours written in different units. Another is the present cabin temperature, which the drop does not use. Ask of each whether the relationship needs it. And note what the answer means. The cabin cools slowly because of what it is made of, which buys hours now and costs the same energy to reheat later.

ASK 7W

Estimate the temperature drop over three hours.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

it takes energy to warm a mass, and the same energy leaves when it cools

VERDICT 86W

A lumped estimate treats the cabin and everything in it as 1 thermal mass of about 12 million joules per kelvin. Energy lost divided by heat capacity gives the temperature drop, and over three hours the answer is small — which is the useful part. Thermal inertia is why the cabin cools slowly rather than instantly when the power goes, and it buys the crew hours to decide. It is also why reheating later is expensive: the same heat capacity has to be filled back up.

TAKEAWAY 12W

Thermal inertia can make temperature change slowly even when power is lost.

7.2 Choose the heat-transfer mechanism

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A PERSON · PROTOCOL

SCENE 32W

The cabin is cooling and the crew wants to do something about it. What can be done depends on how the heat is actually leaving, so the room names the pathway first.

GUIDE 55W

Four situations, and four ways heat moves. Pair them by asking what carries the energy: touching, moving air, or nothing at all. One of these needs no material medium. The pairing decides what a fix looks like, because insulation, circulation and reflective surfaces each act on a different pathway. Naming the wrong one buys nothing.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

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Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

heat moves by contact, by moving fluid, or by radiation

VERDICT 81W

Each heat-transfer pathway suggests a different intervention. Direct contact between people transfers heat by conduction. Moving cabin air transports heat by convection. Infrared exchange between warm objects and a cold wall is radiation, which needs no material medium. A metal strut carrying heat into colder structure is conduction through a solid. Naming the pathway matters because insulation, circulation, reflective surfaces, and thermal isolation act on different mechanisms. A useful fix starts by identifying which pathway is actually carrying the energy away.

TAKEAWAY 9W

Identifying the pathway reveals which intervention can reduce loss.

7.3 Keep the cabin survivable

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 33W

The cabin is at four degrees and falling, condensation is forming on the walls, and the crew is sleeping in their suits. Heating everything is not affordable on the energy that is left.

GUIDE 63W

All four options are ways to spend a fixed energy budget. Ask of each whether it slows the loss or replaces energy already gone. Insulation does the first and cannot do the second. Then ask where cold costs most. Cold batteries deliver less usable energy, so a thermal problem becomes a power problem. The crew is asleep in their suits at four degrees.

ASK 15W

Four degrees and falling, on a fixed energy budget. What do you spend it on?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

slowing a loss and replacing it are different things · newton's laws and free-body reasoning — taken as read

VERDICT 78W

Insulation slows heat loss; it does not replace energy that has already left. That makes insulation valuable when the power budget is fixed. Put it around the crew zone and critical batteries, where cold has the largest consequence. Cold batteries also deliver less usable energy, so a thermal problem can become a power problem. Heating the whole vehicle spends scarce energy on empty volume and exterior surface. Thermal survival is therefore an allocation across space, time, and consequence.

TAKEAWAY 11W

Thermal survival is an allocation problem across space, time, and consequence.

7.4 Cabin pressure, through the work period

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A PERSON · HOLD

SCENE 44W

The crew are working in the middle module and the cabin has to hold pressure while they do. A sample airlock cycles, the scrubber comes on line, and the leak that has been on the log for a week is still on the log.

GUIDE 48W

Hold the cabin pressure inside the band on the console. The band narrows through the work period, because the crew are moving toward the module with the suspect seal and a margin that was comfortable at the start is not comfortable there. The makeup valve is your control.

ASK 10W

Hold the cabin inside the band through the work period.

BEHIND THE BUTTON 125W

Why a leak is not a step. A leak is a rate: the cabin loses gas for as long as the hole is open, so the pressure keeps falling and does not settle anywhere. That is why a cabin with a small leak is held rather than corrected.

Why the airlock and the scrubber are different in kind. Cycling the airlock takes a volume of gas out once. The scrubber changes what the cabin is doing continuously. Both show up on the same gauge and want different answers.

Why the margin matters more later. Everything the crew are doing is reversible except being in the wrong module when the pressure goes. The band narrows because the cost of the same excursion rises as they move.

TAKES AS READ

a sealed volume loses pressure if gas escapes from it

VERDICT 154W

A leak is a rate and everything about handling one follows from that. The cabin loses gas for as long as the hole is open, so the pressure keeps falling and never settles; a valve opened once and closed again buys a few minutes and changes nothing. What works is setting the makeup flow to match the loss and leaving it there, which is also what makes the leak measurable — the flow you need to hold steady *is* the leak rate, and that number is what the ground needs to decide whether the module can stay in use. The airlock is the opposite case: cycling it removes a fixed volume once, so it wants a correction rather than a new setting. Telling those two apart on one gauge is the skill, and the band narrows because the crew are moving toward the module with the suspect seal, where the same excursion costs more.

TAKEAWAY 17W

A leak is a rate, so it is answered by a setting rather than by a correction.

7.5 What holds it in the curve

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · CHOICE

SCENE 35W

The free-return path swings the spacecraft round the Moon at 2,500 kilometres altitude. Somebody on the loop has asked why it does not simply fall onto the surface, and the room owes them an answer.

GUIDE 61W

All four options offer something that holds the spacecraft up. Ask of each whether it names a real force. Two of them invent one: an outward push, and lift with nothing to push against. Gravity is the only force acting here, and it points inward. So the question is what the sideways speed does to a path that is already falling.

ASK 14W

Why does the spacecraft need any speed at all to stay in that orbit?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an object moving in a circle is accelerating even at constant speed · kinematics — taken as read · newton's laws and free-body reasoning — taken as read

VERDICT 76W

Circular motion requires acceleration toward the centre. Around the Moon, gravity supplies that inward acceleration. The spacecraft is therefore falling during the entire orbit. Its sideways speed makes the falling path curve around the Moon instead of into the surface. Too little speed makes the path intersect the Moon. Too much opens the path outward. No engine must hold the spacecraft up. The crew feel weightless because there is no support force opposing the free fall.

TAKEAWAY 18W

A curved path needs a force toward the centre, and orbital motion is that force being exactly enough.

END OF THE SHIFT 17W

Use heat capacity, conduction, convection, radiation, energy balance as an evidence chain rather than as isolated facts.

The Air Problem

PLAN CARD 109W

58 hours in. Carbon dioxide is climbing in a cabin built for two and holding three. The life support engineer, Petra Novak, has been on the loop with the crew for an hour, and the canisters that fit are aboard the module they cannot use. The scrubber fan still draws its rated current, which Brooks refuses to accept as good news. Today you find out why the air is not being cleaned, work out how fast three people fill a cabin, and build something out of what is aboard. Hale has started making small mistakes reading numbers back. That is the first symptom, and he cannot notice it himself.

BRIEFING 24W

Carbon dioxide removal is failing, and pressure readings differ between compartments. The player must connect gas laws, partial pressure, fluid flow, and filter resistance.

WHAT YOU DECIDE 16W

Develop a safe temporary air-cleaning and circulation strategy using only validated physical principles and monitored limits.

8.1 Diagnose the air system

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHAIN

SCENE 32W

Cabin carbon dioxide keeps rising. The scrubber fan sounds normal and draws its usual current, but airflow is low. Pressure readings are available on both sides of the filter and scrubber bed.

GUIDE 56W

Build the path the air takes, in order, and then name the link where it becomes limiting. The fan sounds normal and draws its usual current, which tells you the fan is working. Pressure on both sides of the filter and the bed is what locates the restriction, so read those rather than trusting the sound.

ASK 7W

Where does the air-transfer chain become limiting?

BEHIND THE BUTTON 151W

Why a working fan and low flow means a restriction. A fan draws current according to how much air it is actually moving. A fan meeting a blockage moves less air, works less hard, and sounds entirely normal — which is why the current reading is evidence that the fan is fine and not that the system is.

What a pressure difference across a component tells you. Flow through a restriction costs pressure. A large drop across the filter with a small one across the bed says the filter is the restriction, and the reverse says the bed is. That is the measurement that separates two very different repairs.

Why carbon dioxide is the clock. It rises whenever removal is slower than production, and the crew's symptoms begin well before the level is dangerous. The chain has to be diagnosed while everyone can still think clearly, which is the real deadline.

TAKES AS READ

a running motor does not guarantee useful flow · air must pass through every required link in a closed loop

VERDICT 81W

The fan supplies the pressure difference that can move air, but every downstream restriction decides how much flow that pressure can produce. Normal motor current therefore proves only that the fan is powered. It does not prove that cabin air reaches the sorbent. A large pressure drop across one link, paired with low total airflow, identifies the bottleneck. Because every carbon dioxide molecule must cross that link before reaching the scrubber, the whole loop performs at the rate that restriction allows.

TAKEAWAY 17W

A transport chain is limited by the required link that most restricts the quantity moving through it.

8.2 CO₂ production scale

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · BALLPARK

SCENE 32W

3 crew members each exhale roughly 20 litres of carbon dioxide an hour. The workshop needs to know what an improvised scrubber has to keep up with over the next six hours.

GUIDE 62W

Five numbers, and two of them belong to a different question: a whole day, and a figure by mass rather than by volume. Ask of each whether it is a rate, a count, or a duration. What comes out is the minimum a scrubber has to keep up with. Anything slower than that only slows the rise rather than holding it steady.

ASK 8W

Estimate the carbon dioxide produced in six hours.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate multiplied by a time is a total

VERDICT 80W

Source rate sets the minimum removal rate. 3 people at 20 litres per hour each produce 60 litres per hour together. Over 6 hours that becomes 360 litres. A scrubber that removes carbon dioxide more slowly than the crew produces it cannot hold the concentration steady. It only slows the rise. The estimate does not need high precision. It needs to tell the workshop whether the improvised system is large enough to matter and how much capacity it must provide.

TAKEAWAY 7W

Source rate sets the minimum removal requirement.

8.3 Restore breathable air

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 36W

Carbon dioxide is at 13 millimetres of mercury and still rising. The limit is 15. Spare canisters are square, the socket is round, and only tape, a suit hose, cloth, and a flight-plan cover are available.

GUIDE 60W

All four options describe something the fix could be judged on. Ask of each whether it changes how much air actually crosses the sorbent. Carbon dioxide is only removed from air that goes through, not past. A big canister with a gap around it does worse than a small one sealed tight. The fan makes the same noise either way.

ASK 16W

Tape, a suit hose, a sock and a flight-plan cover. What must the fix get right?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

air, like water, takes the easiest path it is offered

VERDICT 78W

The fix needs both enough sorbent and a sealed flow path through it. Air that slips around the canister reaches the cabin unchanged. That bypass can look deceptively normal because the fan still runs and makes noise. Carbon dioxide is removed only from air that actually crosses the sorbent. A large canister with a gap around it can therefore perform worse than a smaller one with a tight seal. Flow path and chemical capacity have to work together.

TAKEAWAY 11W

A life-support fix must manage both chemical uptake and fluid flow.

8.4 Small burn, large consequence

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · BALLPARK

SCENE 29W

A 30,000 kg spacecraft takes 6,000 N of thrust for 20 seconds. The flight-dynamics officer needs the resulting acceleration and change in speed before the correction burn is approved.

GUIDE 61W

Five numbers, and two of them belong to other questions. Surface gravity and specific impulse are not in this chain. Ask of each whether it is a force, a mass, or a time. And note the scale of the answer. Four metres per second looks like nothing beside an orbital speed. Days out, it moves the arrival by hundreds of kilometres.

ASK 5W

Estimate the change in speed.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

force applied over a time is an impulse · newton's laws and free-body reasoning — taken as read

VERDICT 79W

Force divided by mass gives an acceleration of 0.20 metres per second squared. Over 20 seconds, that produces a 4 metre-per-second change in speed. The same result comes from impulse divided by mass. Four metres per second looks tiny beside an orbital speed measured in kilometres per second. Applied days before arrival, however, it shifts the encounter by hundreds of kilometres. A late correction must produce the same positional change in far less time, so it costs much more.

TAKEAWAY 10W

Orbital corrections often rely on small velocity changes applied early.

END OF THE SHIFT 19W

Use ideal gases, partial pressure, flow resistance, pressure difference, filtration as an evidence chain rather than as isolated facts.

Communication Fades

BEFORE THE SHIFT — CANVASS

Which consoles actually saw the transient

Something moved on the telemetry at 04:12 and only one console flagged it. Ask along the tiers until enough answers have come back to say whether the spacecraft did something or one string glitched, because those two decide whether anybody wakes the crew.

PLAN CARD 103W

66 hours in, the link went weak at the moment the vehicle turned — down 12 decibels, and Hale is answering calls he can barely hear. The communications lead, Naomi Ito, has the link budget open and her network officer is arguing for a different term in it. Ito wants the pointing checked first and he wants the receiver. This shift you find the cause and get the link back. Losing this costs two things at once, because the signal is also a navigation measurement — the room goes deaf and half blind in the same minute, with a burn still to fly.

BRIEFING 25W

The spacecraft signal weakens and shifts in frequency as geometry changes. The team must separate transmitter failure from antenna pointing, range loss, and Doppler effects.

WHAT YOU DECIDE 14W

Restore a reliable low-bandwidth link and use the signal itself as a navigation measurement.

9.1 Why did the signal fade?

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · CONTROL

SCENE 36W

Voice and telemetry are down about 12 decibels at 2 ground stations. Spacecraft transmitter output is nominal. The antenna solution, redundant transmitter chain, and a backup receiver path can each be tested without committing the mission.

GUIDE 71W

The number you are watching is the received signal strength, currently about twelve decibels low at both stations. Each test swaps or corrects one part of the path without committing the mission, and a response counts only if it is larger than the noise on the link. Two stations failing together already narrows what can be common to both. Run one test, undo it, and name the part the fade follows.

ASK 11W

Which reversible change makes the 12 dB fade appear and disappear?

BEHIND THE BUTTON 179W

Why two stations at once is the clue. The two ground stations are on opposite sides of the Earth, with separate receivers, separate crews and separate weather. A fault in one of those cannot show up at the other, so anything that explains both has to be common to both — which means the spacecraft end, or the geometry between them.

What twelve decibels means. Decibels are a ratio on a logarithmic scale: three of them is about half the power, ten is a tenth, and twelve is about one sixteenth of the power arriving. That is the difference between voice you can work with and voice you cannot, which is why the number matters more than it sounds.

Why every test is reversible here. There is a crew aboard, so a test that cannot be undone is not a test but a decision. The pointing correction, the redundant transmitter chain and the backup receiver path were chosen because each can be tried and put back, which is what makes the reversal half of the method available at all.

TAKES AS READ

decibels add gains and losses on a link budget · a causal test changes one condition while the others stay fixed · kinematics — taken as read

VERDICT 79W

Several things can weaken a radio link, but a causal test asks whether the signal follows one controlled change. Swapping to a backup ground receiver changes almost nothing, so the fade is not local to one receiver. Swapping transmitter chains at the same output also changes almost nothing. Correcting the antenna pointing restores nearly all the missing power. Reversing that pointing change brings the loss back. That two-way response ties the fade to antenna geometry rather than to coincidence.

TAKEAWAY 18W

A reversible one-variable test can separate a causal fault from conditions that merely occur at the same time.

9.2 Wavelength of the radio link

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A PERSON · BALLPARK

SCENE 33W

The communications link is fading, and the antenna team needs the carrier wavelength for its pointing model. The spacecraft transmits at 2.0 gigahertz, while received power is being tracked at the ground stations.

GUIDE 61W

Five numbers, and three of them are traps. One is the carrier misread as megahertz. One is the speed of sound. One is the observed loss. Ask of each whether it belongs to this wave at all. And note what the answer feeds: the beam width scales with wavelength over aperture, so this number sets how forgiving the pointing can be.

ASK 3W

Estimate the wavelength.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

frequency and wavelength are two descriptions of one wave · kinematics — taken as read

VERDICT 78W

Wavelength is wave speed divided by frequency. At 2.0 gigahertz, the radio wavelength is about 0.15 metre. That number also sets the scale of the antenna beam. For a fixed dish size, beam width is roughly proportional to wavelength divided by aperture. A longer wavelength gives a wider, more forgiving beam. A shorter wavelength gives a narrower beam and demands better pointing. The calculation is simple, but it feeds directly into the pointing tolerance used for the link.

TAKEAWAY 11W

Frequency and wavelength are reciprocal descriptions of the same propagating wave.

9.3 Recover the weak link

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · SEQUENCE

SCENE 30W

The carrier is down 12 decibels and drifting in frequency. The crew is answering calls it can barely hear, and the next navigation update depends on getting the link back.

GUIDE 60W

All four steps will be taken, so ask which buys the most for the least. Pointing is the largest term in the link budget and costs nothing but attitude. Narrowing the bandwidth buys signal without touching the spacecraft. And one of these four is not a recovery action at all: it is what the recovered data are then used for.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a slower signal needs less power to be read · waves: frequency, wavelength, speed — taken as read

VERDICT 78W

Recovery combines three disciplines in a fixed order, cheapest and largest first.

Pointing is the biggest term in the link budget and costs nothing but attitude, so geometry is checked before anything else. Reducing the data rate and narrowing the receiver bandwidth buys signal-to-noise without touching the spacecraft.

Coordinating the large ground antennas with predicted Doppler puts the receiver where the signal actually is. And the range and Doppler recovered along the way feed straight back into navigation.

TAKEAWAY 8W

Communication recovery joins wave physics, control, and estimation.

9.4 What failed: the spacecraft or the sensors?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CALLBACK · TRACE

SCENE 37W

Nine minutes after the bang, three digital cabin-pressure channels fall together. A mechanical gauge, an acoustic leak monitor, and the sensor-electronics page are also live. The flight director wants one diagnosis before the crew starts opening panels.

GUIDE 58W

Open each pressure channel to see what it depends on. Keep the ones that would still stand if the sensor electronics were at fault, and untick the rest. Then name the shared source that moved the others. Three channels falling together is not three measurements agreeing; it is one measurement repeated three times if they share a supply.

ASK 14W

Which pressure evidence survives the dependency check, and what shared source moved the rest?

BEHIND THE BUTTON 155W

Why three digital channels can be one measurement. Redundant transducers usually share a power supply, a reference voltage and a signal conditioning card. Any of those failing moves all three readings the same way at the same moment, which is exactly what a real depressurisation looks like on the display.

What the other two instruments are for. A mechanical gauge has no electronics between the cabin and the needle. An acoustic monitor listens for escaping gas rather than measuring pressure at all. Neither depends on the suspect chain, which is why they carry more weight here than any number of digital channels.

Why the diagnosis has to come before the panels. Opening panels in a cabin that is genuinely losing air spends time the crew may not have. Sending them to chase an instrumentation fault costs the same hour with no leak to find. The flight director needs one answer, not a list of possibilities.

TAKES AS READ

several instruments can share one reference circuit · an independent measurement does not inherit that shared failure

VERDICT 74W

Agreement is strong evidence only when the measurements can fail independently. The three digital pressure channels all inherit the same reference circuit. When that reference shifts, all three move together even if cabin pressure does not. The mechanical gauge has a different physical path, so its steady reading still counts. The acoustic monitor also finds no leak. Tracing dependencies turns three alarming numbers into one electrical fault and preserves the evidence that still stands.

TAKEAWAY 16W

Several agreeing readouts count as one piece of evidence when they inherit the same physical reference.

END OF THE SHIFT 17W

Use waves, frequency, wavelength, inverse-square behavior, Doppler effect as an evidence chain rather than as isolated facts.

A Blind Maneuver

PLAN CARD 110W

74 hours in, and this is the quiet shift. There is no guidance platform and no star tracker, so Ruiz will aim the burn by eye through a window reticle while Sato holds a stopwatch. Reyes has spent the shift separating what actually moves the spacecraft's aim from what only appears to. This shift you decide how many sightings make an alignment trustworthy. A tenth of a degree is invisible in that window and hundreds of kilometres at the far end. Some of what Ruiz sees is real motion. Some is the window, the lighting, or his own eye at four degrees. Each sighting costs him 20 minutes of cold.

BRIEFING 28W

The crew must align for an engine burn using a window view, a reticle, and imperfect star references. The player must connect optics, angular measurement, and reference frames.

WHAT YOU DECIDE 11W

Design a manual alignment procedure whose dominant angular errors are bounded.

10.1 What shifts the apparent direction?

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE ·
PROTOCOL

SCENE 35W

The crew will aim the next maneuver through a window reticle. Before that view is trusted, the room compares eye position, focus, target parallax, and spacecraft rotation so each apparent shift has a physical interpretation.

GUIDE 58W

Four apparent shifts, and four causes. Pair them by asking what actually moved. The eye? The optics? The viewpoint? The vehicle? Two of them change what you see without changing the line of sight. The crew will aim the next manoeuvre through this reticle. A shift blamed on the wrong cause becomes a burn in the wrong direction.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

what you see through an eyepiece depends on where your eye is

VERDICT 87W

Manual navigation works only if instrument geometry is separated from spacecraft motion. Move your eye relative to a nearby reticle and the apparent alignment changes. That is instrument parallax. Changing focus affects sharpness but not the true line of sight. A nearby object shifting against distant stars as the viewpoint changes is geometric parallax, which carries distance information. If the star field

appears to rotate while the stars themselves remain fixed, the spacecraft body has changed orientation. The same visual shift can therefore have very different causes.

TAKEAWAY 9W

Manual navigation requires separating instrument geometry from spacecraft motion.

10.2 Reading the trace instead of the number

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · PROTOCOL

SCENE 36W

The plot board has 90 minutes of range against time on it, and the room keeps quoting the latest value out loud. The shape of the line has more in it than the last point does.

GUIDE 65W

Four features of one plotted line, and four things they could mean. Pair them by asking what changes and what does not. A slope is a rate. A change of slope means something acted. Scatter around a smooth line is the instrument. A step with the same slope after it is bookkeeping, and that is the one the room will explain with an invented manoeuvre.

ASK 15W

Match each feature of the plotted trace to what it tells you about the motion.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

the slope of a distance-time graph is a speed · kinematics — taken as read

VERDICT 83W

A distance-time graph contains more information than its newest point. The slope is the rate of change, so it gives speed. A change in slope is acceleration and signals that a force acted. Random scatter around a smooth trend is measurement noise rather than real motion. A sudden step followed by the same slope is different. It suggests an offset in calibration, units, or bookkeeping. Reading the shape of the trace prevents the room from inventing a maneuver to explain an instrument change.

TAKEAWAY 17W

A graph carries rate and change of rate, neither of which is visible in the latest reading.

10.3 How far it drifts while nobody is steering

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · BALLPARK

SCENE 31W

Between burns the spacecraft is unpowered. A residual acceleration of 0.004 metres per second squared acts for 90 seconds. The navigation team needs the resulting drift before deciding whether it matters.

GUIDE 67W

Four numbers, and one of them is surface gravity, offered for scale rather than for use. Another is the one-half that belongs in the relationship. Ask of each whether it is an acceleration, a time, or a constant. And note the square on the time. Double the coast and the drift is four times larger, which is why the coast length is the thing to argue about.

ASK 7W

Estimate the drift over the 90 seconds.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

an object under constant acceleration covers half the acceleration times the time squared · kinematics — taken as read · newton's laws and free-body reasoning — taken as read

VERDICT 70W

For constant acceleration from rest, displacement grows as one-half the acceleration times time squared. Here the result is about 16 metres. The square on time is what makes the effect important. Double the coast time and the drift becomes four times larger. A six-minute coast therefore produces far more than twice the displacement. The same mathematics describes free fall and projectile motion whenever one approximately constant acceleration dominates the motion.

TAKEAWAY 12W

Under a constant acceleration, displacement grows with the square of the time.

10.4 Read rotational motion

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · CALLBACK ·
PROTOCOL

SCENE 39W

The main guidance platform is unavailable and the crew is about to fly the vehicle by hand. Before anybody touches a thruster the room states the rotational mechanics plainly, because each one has a consequence the crew will feel.

GUIDE 67W

Four changes on the left, and four consequences on the right. Pair them by asking what each change does to the two terms that matter: the torque, and the resistance to turning. One of the four adds no external torque at all, which is what makes it different from the others. The crew is about to fly this by hand and will feel every one of them.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a torque is what makes something start or stop turning · newton's laws and free-body reasoning — taken as read

VERDICT 82W

Attitude control is Newtonian mechanics about a centre of mass. Torque grows with force and with the lever arm from that centre. The same torque produces less angular acceleration when the moment of inertia is larger. Equal and opposite torques applied for equal times cancel the net angular impulse. Internal mass motion can change how fast different parts rotate, but it cannot change the vehicle's total

angular momentum without an external torque. Those four rules set the logic of every manual turn.

TAKEAWAY 10W

Attitude control is Newtonian mechanics around a center of mass.

END OF THE SHIFT 18W

Use lenses, angular measurement, parallax, reference frames, error propagation as an evidence chain rather than as isolated facts.

Crossing the Atmosphere

PLAN CARD 111W

82 hours in, and nothing goes wrong this shift, which is its own kind of unnerving. The capsule will arrive at about 11 kilometres a second into a corridor roughly a degree wide, and Erika Lindqvist, integrating the flight dynamics, wants the whole room to understand what that means before anybody is asked to vote on it. This shift you follow where the energy goes, put a number on it, and decide how to protect the corridor. Too shallow and the capsule skips back out with no consumables to come round again. Too steep and the shield takes everything in seconds. The same physics keeps them alive and would kill them.

BRIEFING 29W

The return corridor is narrow: too steep causes extreme heating and deceleration; too shallow risks skipping back into space. The player must integrate drag, energy conversion, and atmospheric density.

WHAT YOU DECIDE 12W

Choose a corridor and monitoring strategy robust to atmospheric and navigation uncertainty.

11.1 Where orbital energy goes

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · SEQUENCE

SCENE 35W

The capsule reaches the atmosphere carrying enormous kinetic energy. It must arrive at the ocean with almost none of that energy left. The entry team traces where the energy goes as drag begins to act.

GUIDE 64W

These four are one chain, and reentry is an energy problem before it is a temperature problem. Ask of each card what has to be true first. There is no atmospheric braking before there is atmosphere. Energy cannot become heat until it has left the vehicle's motion. The last card is where the shield and the trajectory decide the rates rather than the total.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

energy is not destroyed, it is moved somewhere else · newton's laws and free-body reasoning — taken as read

VERDICT 81W

Reentry is an energy-transfer problem before it is a temperature problem. The capsule arrives with large kinetic and gravitational energy. Atmospheric drag does negative work and removes mechanical energy from the vehicle. That energy does not disappear. Much of it becomes thermal energy in the shocked gas and wake, while some enters the shield and structure. Speed and altitude fall as the transfer continues. The trajectory and heat shield control how quickly the energy leaves and where the heating is concentrated.

TAKEAWAY 11W

Reentry is an energy-dissipation problem constrained by human and material limits.

11.2 Kinetic-energy scale

THERMAL & LIFE SUPPORT • LIFE SUPPORT LAB • A A PLACE • BALLPARK

SCENE 30W

A 5,000 kg capsule returning from the Moon arrives at about 11,000 metres per second. The thermal protection team needs the kinetic energy before it can argue about heat load.

GUIDE 60W

Five numbers, and two of them belong to other entries: the speed from low orbit, and surface gravity. Ask of each whether this energy depends on it. And note the square on the speed. Forty per cent more speed is nearly twice the energy at the same mass. That is the whole difference between the two entries on the list.

ASK 7W

Estimate the kinetic energy to be dissipated.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

kinetic energy depends on mass and on speed • newton's laws and free-body reasoning — taken as read

VERDICT 77W

Kinetic energy is one-half the mass times speed squared. The squared speed is the dominant feature of lunar return. Raising entry speed by about 40% nearly doubles the kinetic energy for the same mass. That extra energy must still be removed before landing. The heat shield therefore faces a much harder job even if the capsule itself is unchanged. Comparing speeds without squaring them would badly understate the difference between low-Earth-orbit entry and return from the Moon.

TAKEAWAY 8W

Speed dominates kinetic energy because it enters squared.

11.3 Protect the entry corridor

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CLOUD

SCENE 30W

The safe entry corridor runs from 5.3 to 7.7 degrees. The current solution is centered at 6.2 degrees with a one-sigma width of 0.5 degrees. Two actions remain before commitment.

GUIDE 67W

The band is every entry angle the tracking permits: centred at 6.2 degrees with half a degree of one-sigma width, against a corridor from 5.3 to 7.7. Two actions remain, and they do different things — one moves where the band sits, the other changes how wide it is. Apply what you would commit to, watch both numbers, then say when the solution is safe to fly.

ASK 27W

Can you make the entry solution robust without confusing a better center with a narrower uncertainty? Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 172W

Why the width is the thing. A trajectory outside the corridor is a vehicle that either skips back out or comes in too steep, and the band is the set of trajectories the tracking cannot rule out. Being centred perfectly with a wide band still leaves some of those outside — and a crew flies one draw from that band, not its middle.

Two different actions. A targeting correction moves the nominal, which slides the whole band along the corridor. A tracking pass adds information, which is the only thing that makes it narrower. Confusing them is the classic error in this room: the number on the display gets better and the risk does not.

Why 'when is it ready' is part of the question. Every action costs time on a clock the vehicle sets, and the crew cannot wait for an arbitrarily tight solution. So the decision is not merely what to do but when the band is inside the corridor by enough that the next hour is better spent elsewhere.

TAKES AS READ

an uncertainty band describes a range of plausible states · a narrower distribution puts more probability near its center

VERDICT 84W

The nominal entry angle is only the center of a distribution. Moving that center does not make the state better known; the whole uncertainty cloud moves with it. Extra independent tracking does something different by narrowing the cloud. Here neither action alone reaches the required probability inside the safe corridor. Centering removes the bias, while added tracking reduces the spread. Together they move nearly all plausible entry states away from both boundaries. That is what turns a good nominal trajectory into a robust one.

TAKEAWAY 17W

Moving an estimate changes its center; new information changes its width, and safety can depend on both.

11.4 Torque from a thruster

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CALLBACK · BALLPARK

SCENE 31W

A 200 N thruster fires at right angles, 3 metres from the centre of mass. The crew needs the torque before it can predict how fast the vehicle starts to turn.

GUIDE 61W

Five numbers, and three of them belong elsewhere: the vehicle mass, the docking ring, and a cosine. Ask of each whether the turning effect depends on it. The force is at right angles here, so no angle factor is needed. And note what the crew can actually change. Choosing a different thruster pair changes the lever arm without changing the thruster.

ASK 8W

Estimate the torque about the centre of mass.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a longer lever arm gives the same force more turning effect · newton's laws and free-body reasoning — taken as read · rotational kinematics — taken as read

VERDICT 78W

Torque is force times lever arm times the sine of the angle between them. Here the force is perpendicular, so that sine is one. The calculation is therefore just 200 newtons times 3 metres. The result is 600 newton metres. Move the same thruster closer to the centre of mass and the torque falls in direct proportion. On this vehicle, the crew can choose which thruster pair fires. That choice changes the lever arm without changing the thruster.

TAKEAWAY 11W

Force location matters as much as force magnitude in rotational control.

END OF THE SHIFT 16W

Use kinetic energy, drag, work-energy, heating, deceleration as an evidence chain rather than as isolated facts.

The Structure Is Vibrating

BEFORE THE SHIFT — EVADE

The public affairs officer who wants a line for the networks

He has been at the back of the room since the anomaly wanting something to say, and the room has not yet agreed what happened. Keep out of his way. A description given before the room agrees is a description the room then has to correct on television.

PLAN CARD 100W

90 hours in, and a structural dynamics engineer, Chidi Okoye, caught this 1 — worth saying, because nothing else this week has been caught early. A panel starts shaking whenever a pump runs near one particular speed, and a line behind it has been rubbing for an hour. Mensah has the numbers ready. This shift you decide what is driving it, work out the structure's own natural rhythm, and stop it. The shaking dies away as soon as the pump moves off that speed, which is the part Okoye thinks matters. This is a shift where something is simply fixed.

BRIEFING 26W

A pump and flexible panel produce a growing oscillation near one operating speed. The player must identify resonance and protect the structure without disabling life support.

WHAT YOU DECIDE 15W

Move the system away from resonance and verify that the mitigation works across operating conditions.

12.1 What is driving the vibration?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · CHOICE

SCENE 34W

A structural vibration becomes severe only near 3,200 rpm of reaction-wheel speed. The amplitude falls away above and below that band. Two accelerometers on different parts of the structure record the same narrow response.

GUIDE 63W

Four options, and three facts have to be covered. The response peaks in a narrow band of wheel speed. It falls away either side. Two accelerometers on different parts of the structure see the same thing. Ask of each option how many of the three it explains. A loose sensor would not be confirmed by a second one, and an impact happens once.

ASK 10W

Which explanation fits the frequency, speed dependence, and independent accelerometers?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

every structure has speeds at which it prefers to shake

VERDICT 80W

Resonance is identified by how response changes with forcing frequency. A structure can be strong under static load and still respond strongly near one natural frequency. Here the vibration peaks in a narrow band of wheel speed. Moving away from that band reduces the motion immediately. A second accelerometer sees the same pattern, which argues against one loose sensor. The repeatable relationship between forcing speed and structural response is the key evidence. Large amplitude alone would not identify the cause.

TAKEAWAY 14W

How hard something shakes matters less than what is shaking it and how fast.

12.2 Spring-mass period

ROTATION & STRUCTURES · STRUCTURES TEST BAY · A A PLACE · BALLPARK

SCENE 31W

A mounted component is modeled as a 4 kg mass on a 400 N/m spring. The structural analyst needs its natural period and frequency before nearby equipment operating speeds are approved.

GUIDE 58W

Five numbers, and three belong to other things on the vehicle: a wheel speed, a damper, and a nearby mode. Ask of each whether this component's own oscillation depends on it. Two terms set it: how much mass there is, and how stiff the mount is. And note which way each pushes, because they act in opposite directions.

ASK 4W

Estimate the natural period.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a stiffer spring vibrates faster and a heavier mass vibrates slower · newton's laws and free-body reasoning — taken as read

VERDICT 78W

For a spring-mass system, inertia and stiffness set the natural period. More mass makes the oscillation slower, while greater stiffness makes it faster. With 4 kilograms and 400 newtons per metre, the period is about 0.63 second. That corresponds to roughly 1.6 hertz. The number matters because repeated forcing near a natural frequency can build a large response. Knowing the mode first lets the team keep operating speeds away from it instead of discovering the resonance in flight.

TAKEAWAY 7W

Natural frequency emerges from inertia and stiffness.

12.3 Stop the resonance

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · CHOICE

SCENE 34W

A structural panel oscillates whenever the pump runs near one speed. The amplitude has grown for three cycles. The pump is needed, but its operating speed can still be changed from the control panel.

GUIDE 60W

Resonance needs two things: a structure, and a forcing frequency. All four options act on one of them, or on neither. Ask of each which, and how fast it can be done from where the crew is sitting. The amplitude has already grown for three cycles. So one option's assumption, that the structure survives the passage, is no longer safe.

ASK 7W

What is the safest immediate control-panel action?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a resonance needs both a structure and something driving it · newton's laws and free-body reasoning — taken as read

VERDICT 81W

Resonance needs both a structure and a periodic forcing frequency. In flight, changing the forcing is usually faster than changing the structure. Moving the pump speed away from the measured resonant band therefore reduces the response immediately and reversibly. Added damping is a strong permanent fix, but it requires hardware and access. Driving faster through the band assumes the structure survives the passage. The growing amplitude has already made that assumption unsafe. The immediate action is to move the forcing frequency.

TAKEAWAY 16W

A structural fix should change the dynamics and then demonstrate that the dangerous mode is controlled.

12.4 What the room is calling now

WAVES & COMMUNICATIONS · CAPSULE COMMUNICATIONS · A A PLACE · SPOT

SCENE 43W

Calls come to the loop faster than they can all be answered. The flight director holds one rule about which get answered first, and changes it as the situation does — after the burn, after the leak is confirmed, after the crew wake.

GUIDE 59W

Answer the calls the current rule wants and hold the rest. The rule is called once on the loop and then it stands; nobody repeats it. What is scored is the calls either side of a change, because they are the only ones that show whether the room is working to this rule or to the one before it.

ASK 12W

Answer to the rule in force, and keep listening for the rule.

BEHIND THE BUTTON 92W

Why a flight rule changes. Before a burn everything defers to the burn. Once a leak is confirmed everything defers to the leak. When the crew are asleep the rule is different again, because a call that wakes them costs something a call to the ground does not.

Why the changeover is the dangerous part. A room working to the old rule for two more minutes gives the loop to a call that no longer matters, and the call that does matter waits behind it with nothing to say it was displaced.

TAKES AS READ

a control room answers calls in an order somebody decides

VERDICT 119W

Three rules run across the shift and each wants a different part of the board. Anything affecting the burn, then anything affecting the leak, then anything that would wake the crew. A call can answer two at once, which is what makes the change cost real rather than notional. What the panel scores is the window either side of each change, because most of what arrives is wanted by neither rule and is correctly held by somebody who has read nothing. The room's own version is the two minutes after a leak is confirmed: the rule has changed, and the loop is still being given to the trajectory people because that is who it belonged to a moment ago.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

12.5 How long can the battery last?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · CALLBACK · BALLPARK

SCENE 31W

The fuel cells are gone and the crew is on entry batteries days earlier than any plan allowed for. The flight surgeon wants an endurance number before the next powerdown decision.

GUIDE 68W

Five numbers, and three of them belong to other questions: the bus voltage, the brief peak, and the time to the next burn. Ask of each whether it is an amount of energy or a rate of use. Energy divided by a rate leaves a time. And note what the answer assumes: a flat load, and every last watt-hour available. A cold battery gives less than its rating.

ASK 4W

Estimate the ideal endurance.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

energy is a stock, and power is the rate it is drawn at · circuits: current, voltage, resistance, Ohm's law — taken as read

VERDICT 81W

Energy and power are different things, and the division between them is the whole calculation. An inventory in kilowatt-hours divided by a draw in kilowatts leaves hours. It is an ideal figure. It assumes the load stays flat and that every last watt-hour is available, and neither is true — a cold battery gives less than its rating. It is still the number every other decision this shift gets argued against, so it is worth having early and worth calling approximate.

TAKEAWAY 15W

Power is the rate of energy use; endurance depends on both energy inventory and load.

END OF THE SHIFT 19W

Use simple harmonic motion, natural frequency, resonance, damping, forced oscillation as an evidence chain rather than as isolated facts.

Choose the Return Path

PLAN CARD 96W

98 hours in. The correction Whitaker flew on the third shift has come back with a cost: tracking now says the path he chose is the expensive one, and Carter has not said anything about it, which is somehow worse. Four routes are on the board — fastest, least propellant, coolest entry, best tracking coverage — and once the burn flies most of them close. This shift you find what actually limits this vehicle and pick the path that survives being wrong. What you want is the path with something left when the next thing breaks.

BRIEFING 24W

Several return trajectories trade fuel, time, heating, communications, and life-support consumption. The player must make a multi-constraint physical decision rather than optimize one variable.

WHAT YOU DECIDE 13W

Select a return plan with explicit margins and contingencies for the dominant uncertainties.

13.1 Name the binding constraint

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · PROTOCOL

SCENE 36W

Four return paths are on the mission board. Each one stresses a different subsystem, and the fastest route is not best on every constraint. The flight director asks the room to name what each option spends.

GUIDE 54W

Four return paths, and four subsystems that pay for them. Pair them by asking what each path spends: consumables, propellant, heating, or the crew's own ability to navigate. No path is cheapest on every count. So the useful question is not which is fastest but which constraint each one pushes closest to its limit.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a spacecraft carries a fixed amount of everything it needs

VERDICT 77W

A trajectory is a system choice because its costs land on different subsystems. More flight time spends oxygen, water, scrubber capacity, and electrical energy. A larger correction burn spends propellant and increases execution error. A steeper atmospheric entry raises peak heating and deceleration. A long communications blackout leaves the crew without ground support when navigation is most difficult. No single path minimizes every cost. The useful question is which constraint each path pushes closest to its limit.

TAKEAWAY 12W

The best physical solution is not necessarily the shortest or lowest-fuel solution.

13.2 Select the robust trajectory

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · STRESS

SCENE 35W

Four return paths are on the board: 58, 63, 66, and 71 hours. Their propellant and entry margins differ. The usable consumable endurance is uncertain by about a day, and water is the tightest supply.

GUIDE 58W

Four return paths, and the consumable estimate is uncertain by about a day. Move that estimate to its pessimistic end before choosing, and watch which trajectories go dark. A path that works at the nominal estimate and fails a day short is not a plan. Water is the tightest supply, so it is the one to test against.

ASK 14W

Which trajectory still works when the life-support estimate is wrong in the bad direction?

BEHIND THE BUTTON 143W

Why the pessimistic end rather than the expected one. The endurance figure is an estimate with a spread, not a measurement. Choosing at the middle of it means accepting a real chance of running out, and running out of water on a return trajectory is not recoverable by any later decision.

What the trade actually is. A faster path uses more propellant and leaves less entry margin; a slower one asks more of the consumables. Neither end is safe, which is why the choice is made by testing both against their uncertain quantity rather than by minimising one number.

Why water rather than oxygen or power. It is the supply with the least margin, so it fails first, and every hour added to the trajectory costs it directly. The binding constraint is the one worth stressing; testing against a comfortable supply proves nothing.

TAKES AS READ

a constraint can eliminate an otherwise attractive option · uncertainty should be tested in the direction that makes the plan hardest

VERDICT 84W

A nominal comparison hides which path fails first when an uncertain input moves. The shortest return looks best if time is the only score, but its entry margin is too small. The longest options preserve other resources but depend on life support lasting beyond the pessimistic estimate. The balanced path is not the nominal winner on speed. It is the only candidate that keeps enough entry and propellant margin while also returning before the low-end consumable limit. That is robustness rather than average performance.

TAKEAWAY 20W

A robust choice survives the plausible error in the assumption that matters most, even when another option looks better nominally.

13.3 Commit to the path

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · SEQUENCE

SCENE 31W

The decision has to be made this shift, and once the burn is executed most of the alternatives close. Nobody has yet written down what would show the choice was wrong.

GUIDE 55W

All four of these will happen, and the burn closes most of the alternatives. Ask of each what would be impossible afterwards. Comparing candidates under the nominal case settles nothing, because the nominal case was never in doubt. And an abort trigger defined after the burn is a trigger with nothing left to act on.

ASK 13W

Order the commitment so the abort trigger still has something to act on.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a decision can only be undone while the means to undo it still exist

VERDICT 89W

A plan is a hypothesis about future physical states, and it needs a way to be found wrong in time. The margins are quantified first, because they are what the decision is made of. The candidates are compared under off-nominal conditions rather than nominal ones, because the nominal case is the one that was never in doubt. The abort trigger is named before the burn — defined afterwards it is a trigger with nothing left to act on. Then the margins keep being updated against what the tracking returns.

TAKEAWAY 16W

An abort trigger has to be defined while there is still propellant to act on it.

13.4 Series or parallel?

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · CALLBACK · PROTOCOL

SCENE 33W

Engineers want to reconnect battery modules to buy hours of endurance. The room agrees what each wiring choice does before anything is bolted up, because adding a module can also take capability away.

GUIDE 61W

Four wiring changes, and four consequences. Pair them by asking what stays the same in that arrangement: the voltage across each part, or the current through it. Then ask what one fault does. Adding a module can take capability away, because a single open cell can break a whole series path. This is being bolted up to buy hours of endurance.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a battery has both a voltage and a current it can supply

VERDICT 82W

Circuit topology determines voltage, current sharing, and fault behavior. Equal cells in series add voltage while carrying the same current. One open cell can therefore break the whole series path. Equal cells in parallel keep about the same voltage and share the current. That arrangement can add useful capacity. A resistive parallel branch carries less than its share, forcing the other branches to carry more.

Redundancy only helps when the connection and protection rules prevent one fault from defeating the whole arrangement.

TAKEAWAY 11W

Redundancy is useful only when connection and protection behavior are explicit.

END OF THE SHIFT 17W

Use optimization, constraints, trade spaces, margins, critical path as an evidence chain rather than as isolated facts.

The Last Correction

PLAN CARD 97W

106 hours in, and it is Carter's turn — she has been waiting 4 days for it. Several ground stations have suddenly put the spacecraft ahead of its predicted path. They agree closely, and they are fed from the same timing reference. Whitaker wants the burn now, on the same argument that was correct on the third shift. This shift you decide whether the trajectory really moved. Carter's rule is that a correction is worth making only when it is bigger than the error in knowing it, and she will say so in front of the room.

BRIEFING 25W

Hours before entry, tracking stations disagree slightly. The player must combine measurements with correlated errors and decide whether another burn reduces or increases total risk.

WHAT YOU DECIDE 14W

Decide whether to burn using the expected benefit relative to navigation and propulsion uncertainty.

14.1 Real trajectory error or common clock drift?

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CHOICE

SCENE 36W

Several ground measurements suddenly place the spacecraft ahead of prediction. The range and Doppler solutions agree with one another. An optical star-angle solution and the station timing records are available before any correction burn is commanded.

GUIDE 58W

Four options, and two measurements agree. Ask what they share before treating that as confirmation. Range and Doppler both use the ground timing reference. The optical solution does not, and it matches the prediction instead. So one of these options explains several agreeing numbers as one failure. A correction burn is about to be commanded on the answer.

ASK 10W

Which explanation fits the ranging, angle, clock, and independent-time checks?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

several instruments can be wrong together if they share a part

VERDICT 81W

Agreement matters only when measurements can fail independently. Range and Doppler both use the same ground timing reference, so one clock error can shift them together. Optical star-angle navigation does not use that clock and still matches the predicted direction. The timing reference is then found to be offset. Reprocessing the original observations with corrected timestamps removes most of the apparent trajectory error. No force acted on the spacecraft. The shared dependency created several agreeing numbers that were really one failure.

TAKEAWAY 18W

When several measurements agree, ask whether they agree independently or merely inherit the same clock, calibration, or model.

14.2 Combine independent uncertainties

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A PERSON · BALLPARK

SCENE 38W

The navigation board carries two position errors. 1 is 6 kilometres, the other 8, and they lie along axes at right angles. Sandoval needs a single number for the entry brief, and the burn is called from it.

GUIDE 58W

Five numbers, and three of them are wrong ways to combine the two errors: their sum, their average, and the number of axes. Ask of each whether it respects the geometry. The two errors lie along perpendicular axes, so the answer is larger than either and smaller than their sum. The burn is called from this single figure.

ASK 5W

Estimate the combined two-dimensional uncertainty.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

two errors along different directions do not simply add

VERDICT 78W

Independent errors along perpendicular directions combine in quadrature. Square each width, add the squares, and take the square root. 14 kilometres therefore combine to 10 kilometres. The result is larger than either contribution but smaller than their simple sum. That distinction matters operationally. Adding them directly exaggerates the uncertainty and can motivate an unnecessary burn. Treating the larger term alone understates the uncertainty and can hide a corridor problem. The combination rule preserves the geometry of independent errors.

TAKEAWAY 6W

Independent orthogonal uncertainties combine in quadrature.

14.3 Burn or observe?

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · VALUE

SCENE 34W

Two timing-based tracking solutions disagree, and both depend on the same station clock model. A correction burn is ready. There is time for only a small amount of additional evidence before the decision closes.

GUIDE 56W

Two tracking solutions disagree, and both are computed through the same station clock model. Open each piece of evidence and ask what its result would change about the burn decision. Buy the evidence that could distinguish a real trajectory error from a clock error, not the evidence that would refine a number both solutions already share.

ASK 10W

Which evidence is worth buying before committing the correction burn?

BEHIND THE BUTTON 136W

Why two disagreeing solutions can share a fault. Both are derived from timing measurements referred to the same clock model. A modelling error moves both, and it can move them by different amounts because they use different station geometries — which looks exactly like a genuine disagreement about where the spacecraft is.

What would separate the two explanations. A measurement that does not depend on that clock model: an optical sighting, a different tracking technique, a station whose timing comes from elsewhere. Any of those either confirms the trajectory or exonerates it.

Why the burn makes it urgent. Correcting a trajectory that is not wrong introduces the error you were trying to remove, and spends propellant doing it. The decision closes shortly, which is what makes the choice of evidence a decision rather than a preference.

TAKES AS READ

two measurements that share a clock do not provide independent timing evidence · an observation can be valuable even if it is less precise · newton's laws and free-body reasoning — taken as read

VERDICT 78W

More precision on the same dependency does not resolve a common-mode ambiguity. Reprocessing the same timing data can reduce numerical noise while leaving the clock error untouched. Another range pass tied to that clock has the same weakness. An optical star-angle observation uses different physics and a different reference, so it can tell whether the spacecraft itself is displaced. Under a small budget, that independence is worth more than another precise copy of the evidence already in hand.

TAKEAWAY 17W

The most valuable measurement is the one whose failure mode is independent enough to change the decision.

14.4 Heating at a bad connection

POWER & CIRCUITS · POWER & CIRCUITS BAY · A A PLACE · CALLBACK · BALLPARK

SCENE 33W

A connector in the improvised battery path develops 0.05 ohms while carrying 20 amps. The joint is inside a sealed cabin. The power team needs its heating rate before the hardware is approved.

GUIDE 71W

Five numbers, and three of them belong elsewhere: the bus voltage, the whole cable run, and half the current. Ask of each whether the heating at this joint depends on it. Note the square: at half the current the same joint gives off a quarter of the power. And note where it is. Twenty watts is nothing on a radiator and a great deal in one connector in a sealed cabin.

ASK 7W

Estimate the heating power at the connection.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a resistance carrying current turns electrical power into heat

VERDICT 79W

Resistive heating is current squared times resistance, so the current term dominates. The same connector at half the current would give off a quarter of the power. The resistance sounds like nothing written down. The wattage it makes at this current is what decides whether the junction runs warm or starts a fire. 20 watts is nothing spread over a radiator. It is a great deal in one connector, in a sealed cabin, with no air moving over it.

TAKEAWAY 13W

Small resistances matter when current is large because heating scales with current squared.

END OF THE SHIFT 23W

Use random and systematic error, weighted averages, errors that move together, propagation, decision thresholds as an evidence chain rather than as isolated facts.

Reentry

BEFORE THE SHIFT — TAG

Catch Reyes before the burn briefing

Reyes, the guidance lead, has the updated attitude numbers, and once the burn briefing starts the room is committed to whatever is on the board. The version up there is from before the leak was confirmed, and the burn is calculated against the mass the leak has been changing.

PLAN CARD 124W

A hundred and 18 hours in. The sequence starts in 11 minutes, 4 of them with no communications, and nothing in it can be undone. This shift you take the claims one at a time and decide whether this vehicle is ready. They are not equally strong: a heat shield cold for four days, a battery charged through a path Ferreira invented on the sixth shift, an alignment Ruiz flew by hand against stars, a corridor entered on Carter's independent measurement. Every one of them has to hold at the same moment. After the first event there are no decisions left, only things to watch. Hale, Ruiz and Sato have heard every word of this for five days, and today they say very little.

BRIEFING 22W

The final minutes require every physical model and procedure to agree: attitude, center of mass, velocity, atmosphere, thermal shield, communications, and recovery.

WHAT YOU DECIDE 13W

Conduct a claim-by-claim readiness review and execute entry with predefined triggers and authority.

15.1 Disposition final readiness

MISSION INTEGRATION · FLIGHT DIRECTOR · A A PLACE · ATTEST

SCENE 35W

11 minutes remain before atmospheric entry. Four signed readiness claims sit on the flight director's console. Some have independent evidence behind them and one does not. Only two physical verification checks can still be completed.

GUIDE 55W

Four signed readiness claims, eleven minutes, and two physical checks that can still be completed. Open each claim and read what backs it. Some rest on independent evidence and one does not. Hold what the evidence does not support, and spend the checks where being wrong would matter most once the vehicle is in blackout.

ASK 12W

Which claims deserve the final physical verification before the vehicle enters blackout?

BEHIND THE BUTTON 136W

Why blackout changes the arithmetic. During entry there is no telemetry and no way to intervene. Every claim on that console is a decision that cannot be revisited for several minutes, which is why an unverified claim is worse here than anywhere else in the mission.

What a signature means at this point. Somebody has taken responsibility for a statement. Under time pressure, signatures are given on the basis of what somebody was told, which is how a claim with nothing behind it acquires the same appearance as one with a test behind it.

How to choose between two unbacked claims. By consequence during entry, not by how doubtful each is. A claim whose failure is survivable is worth less verification than one that is not, however much more likely the first is to be wrong.

TAKES AS READ

a signature records responsibility, not physical truth · verification time should be spent according to consequence and existing evidence

VERDICT 83W

Readiness paperwork and physical readiness are different things. A claim backed by independent measurements may deserve less scarce verification time than a critical claim supported only by a log entry. The entry state already has tracking behind it, while the shield configuration changed without a complete record. That is the dangerous gap. Blackout guidance and the known station correction also have direct checks behind them. With only two checks left, verification should target consequence and missing evidence, not whichever signature looks most official.

TAKEAWAY 21W

A signed readiness claim is a record of what someone believes; critical physical states still need evidence proportional to their consequence.

15.2 Execute the final physical chain

MISSION INTEGRATION · FLIGHT DIRECTOR · A PERSON · SEQUENCE

SCENE 32W

Everything now depends on the next 11 minutes, including 4 minutes without communications. The final entry sequence is on the console. Anything needing ground input must happen before the radio goes quiet.

GUIDE 63W

Four minutes with no radio is what orders these four. Ask of each whether it needs ground input, and whether it can be done after the interface. Anything the ground has to approve happens before the blackout. Anything compared against a prediction can only happen once entry has begun. And the record is the only direct test the predicted envelope will ever get.

ASK 10W

Order the entry around the four minutes with no communications.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

you cannot fix from the ground what happens while the radio is quiet

VERDICT 79W

The blackout forces the order of operations. Attitude, configuration, and the approved state estimate are set before atmospheric interface because ground commands will soon be unavailable. The roles of onboard guidance and the ground are also settled before commitment. Deceleration and heating can only be compared with predictions after entry begins. Recovery follows once the high-energy phase is over. The telemetry record is preserved throughout, because this single flight is the only direct test of the predicted entry envelope.

TAKEAWAY 17W

Entry is ordered by the blackout: whatever needs the ground has to be settled before it starts.

15.3 Fund the mission legacy

GUIDANCE & MOTION · GUIDANCE COMPUTER ROOM · A A PLACE · CHOICE

SCENE 36W

The crew is aboard the recovery ship, and the review board meets Monday. Four proposals compete for three funded slots: a tank redesign, a second telemetry path, a full-duration thermal test article, and a simulator upgrade.

GUIDE 62W

All four proposals are worth funding and three slots exist. Ask of each whether it removes a cause or improves the response to one. The same weakness appeared twice this mission: several pressure readouts on one electrical reference, and several tracking products on one clock. Both times, agreement looked stronger than it was. Logs and rehearsal help reconstruct that, not prevent it.

ASK 12W

Which proposal deserves first priority because it removes the repeated physical cause?

BEHIND THE BUTTON 195W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a fix that removes a cause outlasts a fix that improves the response

VERDICT 80W

The mission exposed the same systems weakness twice. Several pressure readouts shared one electrical reference, and several tracking products later shared one clock. In both cases, agreement looked stronger than it really was. A lasting fix removes or isolates those common dependencies. Preserving logs and improving simulation are also valuable because they help reconstruct and rehearse failures. They do not remove the cause. The review should therefore prioritize redesigning common-mode dependencies while keeping the evidence needed to validate that redesign.

TAKEAWAY 17W

The scientific obligation after a crisis is to learn from every discrepancy, not only the successful outcome.

15.4 How much does the cabin cool?

THERMAL & LIFE SUPPORT · LIFE SUPPORT LAB · A A PLACE · CALLBACK · BALLPARK

SCENE 34W

With the systems powered down the cabin is losing about a kilowatt more than it generates. The flight surgeon wants to know how long the crew has before the temperature becomes a medical problem.

GUIDE 64W

Five numbers, and two of them are the same three hours written in different units. Another is the present cabin temperature, which the drop does not use. Ask of each whether the relationship needs it. And note what the answer means. The cabin cools slowly because of what it is made of, which buys hours now and costs the same energy to reheat later.

ASK 7W

Estimate the temperature drop over three hours.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

it takes energy to warm a mass, and the same energy leaves when it cools

VERDICT 86W

A lumped estimate treats the cabin and everything in it as 1 thermal mass of about 12 million joules per kelvin. Energy lost divided by heat capacity gives the temperature drop, and over three hours the answer is small — which is the useful part. Thermal inertia is why the cabin cools slowly rather than instantly when the power goes, and it buys the crew hours to decide. It is also why reheating later is expensive: the same heat capacity has to be filled back up.

TAKEAWAY 12W

Thermal inertia can make temperature change slowly even when power is lost.

END OF THE SHIFT 19W

Use integrated mechanics, thermodynamics, waves, fluids, uncertainty, ethics of command as an evidence chain rather than as isolated facts.

THE CLOSING CARD

Splashdown at 12:07, three good parachutes, and four minutes of blackout that ended with a voice reading its own altitude. The heat shield that had been cold for four days held. The crew were on the recovery deck inside the hour, all three of them walking.

Every claim this room made about that vehicle was written down before entry, and every one of them was still standing afterwards. That is the whole job.

Card text only — no options, boards or answers. 12,276 words of card prose across 60 stops. Generated from themes/bring_them_home by tools/make-cardbook.mjs.